Cross-Spatial Pixel Integration and Cross-Stage Feature Fusion-Based Transformer Network for Remote Sensing Image Super-Resolution

Yuting Lu, Lingtong Min, *Member, IEEE*, Binglu Wang, *Member, IEEE*, Le Zheng, *Senior Member, IEEE*, Xiaoxu Wang, *Member, IEEE*, Yongqiang Zhao, *Member, IEEE*, Le Yang, and Teng Long, *Fellow, IEEE*

***Abstract*— Remote sensing image super-resolution (RSISR) plays a vital role in enhancing spatial details and improving the quality of satellite imagery. Recently, Transformer-based models have shown competitive performance in RSISR. To mitigate the quadratic computational complexity resulting from global self-attention, various methods constrain attention to a local window, enhancing its efficiency. Consequently, the receptive fields in a single attention layer are inadequate, leading to insufficient context modeling. Furthermore, while most transformer-based approaches reuse shallow features through skip connections, relying solely on these connections treats shallow and deep features equally, impeding the model's ability to characterize them. To address these issues, we propose a novel transformer architecture called cross-spatial pixel integration and cross-stage feature fusion-based transformer network (SPIFFNet) for RSISR. Our proposed model effectively enhances context cognition and understanding of the entire image, facilitating efficient integration of cross-stage features. The model incorporates cross-spatial pixel integration attention (CSPIA) to introduce contextual information into a local window, while cross-stage feature fusion attention (CSFFA) adaptively fuses features from the previous stage to improve feature expression in line with the requirements of the current stage. We conducted comprehensive experiments on multiple benchmark datasets, demonstrating the superior performance of our proposed SPIFFNet in terms of both quantitative metrics and visual quality when compared to state-of-the-art methods. Our code is available at https://github.com/Dr-Lyt/SPIFFNet.**

***Index Terms*— Cross-spatial pixel integration, cross-stage feature fusion, remote sensing image super-resolution (RSISR), transformer network.**

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Yuting Lu, Xiaoxu Wang, Yongqiang Zhao, and Le Yang are with the School of Automation, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: lyt1996@mail.nwpu.edu.cn; woyaofly1982@nwpu.edu.cn; zhaoyq@nwpu.edu.cn; nwpuyangle@gmail.com).

Lingtong Min is with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: minlingtong@nwpu.edu.cn).

Binglu Wang, Le Zheng, and Teng Long are with the Radar Research Laboratory, School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: wbl921129@gmail.com; le.zheng.cn@gmail.com; longteng@bit.edu.cn).

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I. Introduction

REMOTE sensing imaging technology is of paramount importance in numerous fields, including environmental monitoring [1], [2], [3], disaster management [4], [5], urban planning [6], [7], and object detection [8], [9], [10]. Therefore, the acquisition of high-resolution (HR) remote sensing images is imperative for the effective implementation and analysis of remote sensing image applications. However, challenges arise due to factors such as sensor noise, optical distortion, and environmental interference, which can significantly degrade the image quality. Image super-resolution (SR) is a typical computer vision task that involves reconstructing HR images from low-resolution (LR) images [11], [12]. The primary objective of SR is to mitigate the detrimental impact of acquisition equipment and environmental factors on remote sensing imaging outcomes, thereby enhancing the resolution of remote sensing images. SR has emerged as a compelling alternative to traditional physical imaging technologies, attracting considerable attention in recent years due to its capability to generate HR remote sensing images effectively [13]. These images encompass a wide range of applications, such as multispectral imagery [14], [15], hyperspectral imagery [16], [17], seismic field data [18], [19], tomographic synthetic aperture radar (TomoSAR) imagery [20], thermal unmanned aerial vehicle (UAV) imagery [21], and more.

Traditional remote sensing image SR (RSISR) methods often rely on interpolation-based techniques, such as bicubic interpolation [22] or Lanczos interpolation. While these methods are simple, they may yield limited performance due to their inability to capture high-frequency details and structural information in the generated images. Recent advancements in deep learning [23] have led to the emergence of convolutional neural networks (CNNs) as powerful tools for various image processing tasks, including RSISR. CNN-based methods [24], [25], [26], [27], [28], [29], [30], [31], [32], [33] have demonstrated promising results in learning complex representations from large datasets. However, despite the success of CNNs, they still possess certain limitations when employed for RSISR. CNNs typically operate locally with fixed receptive fields, which may hinder their ability to effectively capture long-range dependencies. As a result, they may have limited modeling capacity for remote sensing scenes with large spatial extent [34].

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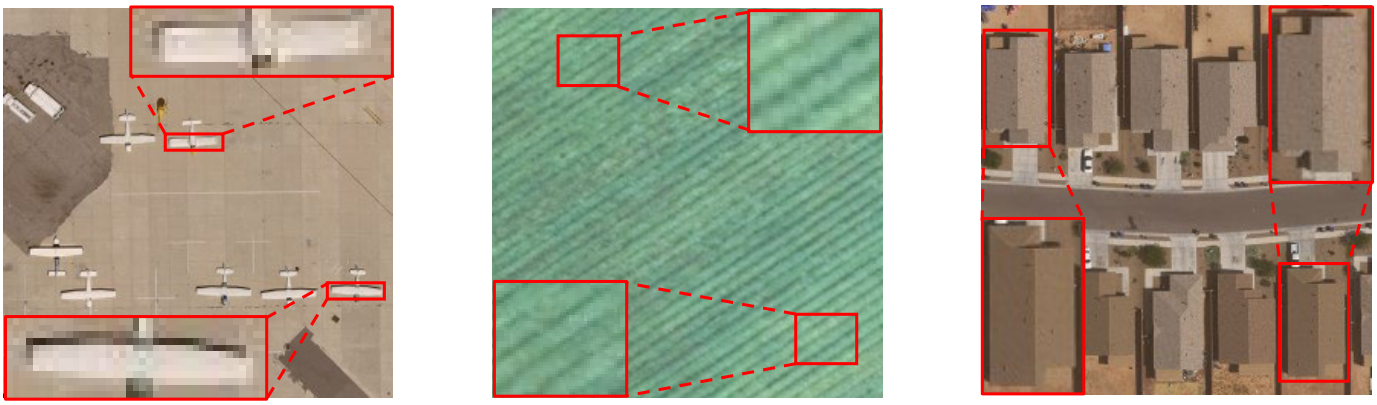


Fig. 1. Examples of similar windows in remote sensing images. These windows are similar in texture, shape, and semantic features but far apart in space.

Transformer-based architectures, initially developed for natural language processing tasks, have emerged as a suitable solution for addressing the limitations of CNNs and have demonstrated impressive performance in diverse computer vision tasks, including image classification [35], [36], [37] and object detection [38], [39], [40]. The vision transformer model [35] utilizes a redundant attention mechanism, resulting in increased computational complexity when applied to HR predictions in RSISR tasks. Recent proposals have explored using self-attention within small spatial regions [41], [42], [43], [44], [45], [46]. However, this approach fails to capture interactions between distant but similar pixels in remote sensing images, where objects and landforms with similar features are far apart, as shown in Fig. 1. Additionally, current transformer-based RSISR methods rely heavily on skip connections to transmit shallow features to deep features. Treating these reused shallow features equally hinders the representational capability of transformers, despite the effectiveness of skip connections in RSISR [47], [48], [49].

To address these limitations, we propose introducing context information into local window self-attention to enhance the model's understanding of image content and calculating cross-covariance of feature channels across stages to generate cross-stage attention map based on both shallow features and deep features to adaptively adjust the channel-wise feature maps at cross-stage of the network. To this end, we propose two components: cross-spatial pixel integration attention (CSPIA) and cross-stage feature fusion attention (CSFFA). CSPIA begins by dividing the image into local and context windows (refer to Fig. 2), then identifies the most matched context window for each local window through similarity calculation, and finally integrates the contextual information into the local window through cross-attention. This effective alignment process establishes meaningful connections between the local window and its context, seamlessly integrating information from the contextual window to enhance the model's overall perception capabilities. The receptive field is substantially enlarged through CSPIA, as illustrated in Fig. 3, thereby significantly improving the model's ability to capture valuable context information from the input image. Additionally, CSFFA enhances feature expression by adaptively integrating cross-stage features. Specifically, CSFFA calculates the

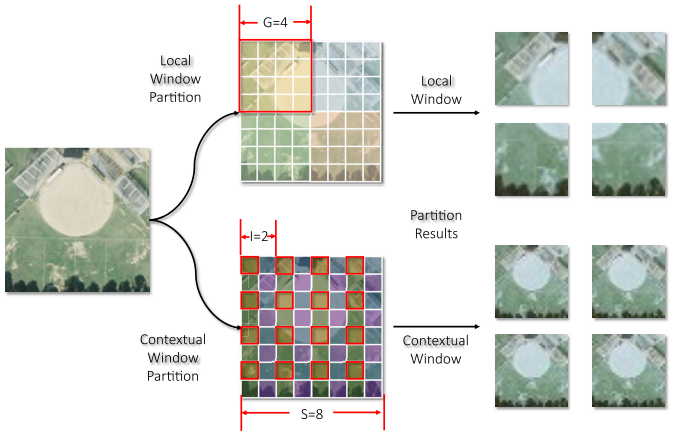


Fig. 2. Diagrams of difference between local window partition and contextual window partition. The pixels in the red box belong to the same window.

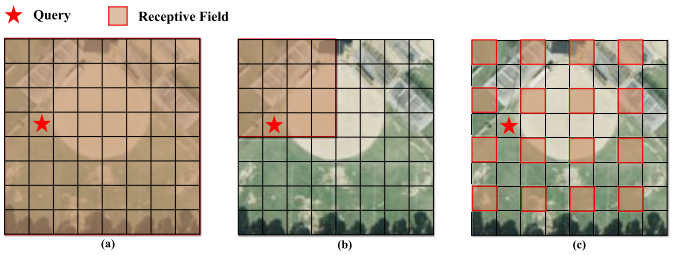


Fig. 3. Comparison of CSPIA with other transformer models. The red star denotes the query, and masks with solid line boundaries denote the regions to which the queries attend. (a) Global attention [50] adopts full attention for all queries. (b) Local attention [42] uses partitioned window attention. (c) CSPIA learns the context information region that is most similar to current query.

cross-covariance of feature channels across stages to generate a cross-stage attention map based on both shallow features and deep features (after the projection of key and query). This adaptive adjustment of channel-wise feature maps at different stages enhances the informative multiscale feature representation ability of the model. Furthermore, to enhance the flow of complementary features and allow subsequent network layers to focus on finer image details, we integrate a feed forward network (FFN) [51] into our model. Based on CSPIA and CSFFA, we develop a transformer network, named cross-spatial pixel integration and cross-stage

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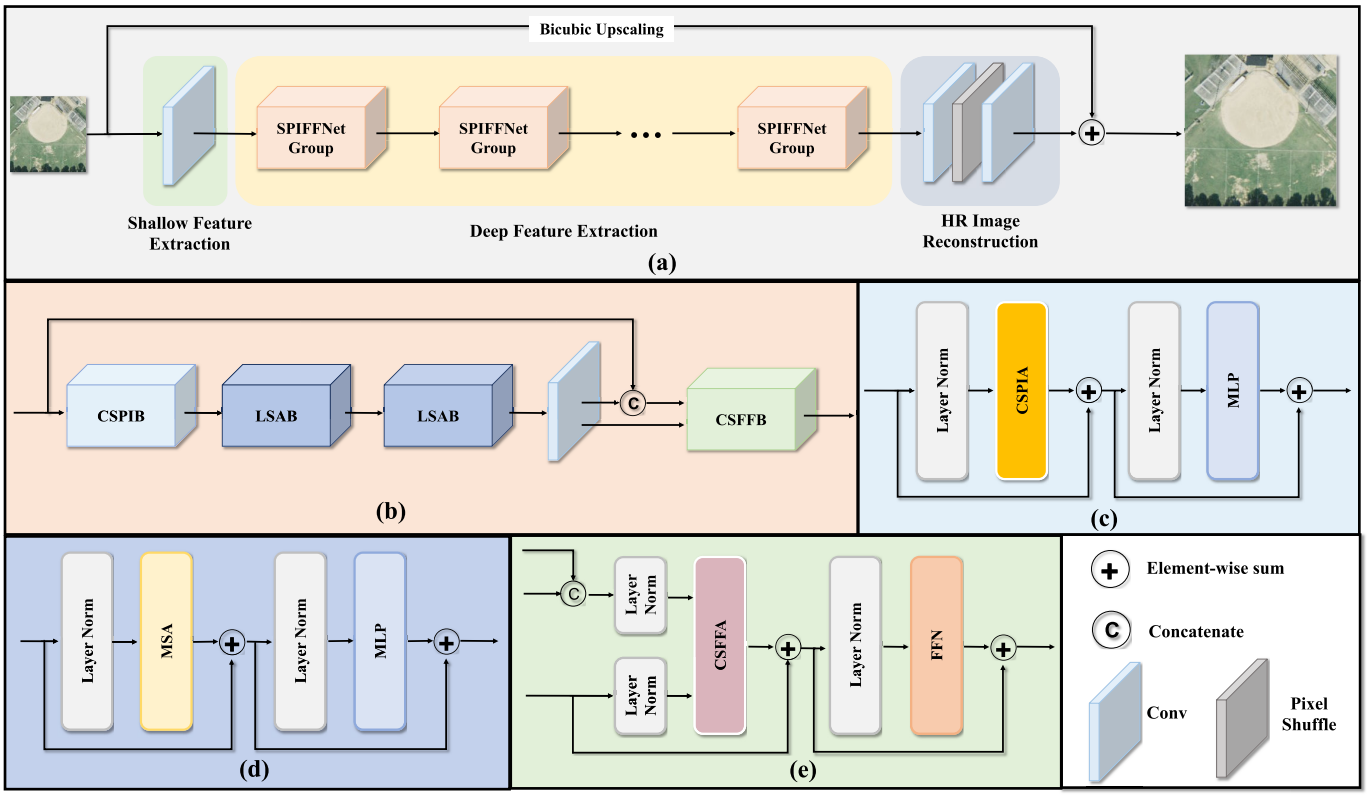


Fig. 4. (a) Architecture of SPIFFNet for HR image restoration. (b) Main components of the model: CSPIB, LSAB, local 3 × 3 convolution, and CSFFB. (c) CSPIB that implements the injection of context information into a local window. (d) LSAB model local-range dependencies in the fine area of a local window. (e) CSFFB performs channel attention across stages for feature fusion.

feature fusion-based transformer network (SPIFFNet), which incorporates cross-spatial pixel integration and cross-stage feature fusion, specifically designed for RSISR as depicted in Fig. 4. Our experimental findings unequivocally demonstrate the superiority of the proposed SPIFFNet model over state-of-the-art methods.

The article presents three key contributions that can be summarized as follows.

1) We propose CSPIA to introduce contextual information into the local window. With the help of CSPIA, the model gains a better understanding of the relationship between ground features and the surrounding environment, thereby enhancing the consistency and accuracy of the RSISR results.
2) We propose CSFFA, which facilitates effective feature representation by modeling the interdependencies among different channels across stages. This mechanism enhances the model's capacity to capture essential image features while suppressing irrelevant ones, thereby producing higher-quality super-resolved images.
3) Building upon CSPIA and CSFFA, we propose SPIFFNet, an SPIFFNet for RSISR. SPIFFNet achieves state-of-the-art performance in terms of objective metrics as well as visual quality.

The rest of this article is organized as follows. Section II presents the related works on SR. Section III introduces the proposed SPIFFNet model, including the CSPIA and the CSFFA. Section IV presents the experimental results and analysis. Finally, Section V concludes this article.

II. Related Work

A. Deep Learning-Based Methods for SR

Deep learning-based SR methods predominantly rely on standard CNNs owing to their robust nonlinear representation capabilities. Typically, these methods approach SR as an image-to-image regression task, with the objective of learning the direct mapping from LR to HR images. SRCNN [52] first uses three convolution layers to map the LR images to HR images. Building upon SRCNN, Kim et al. [53] extended the network depth in their work called DRCN, resulting in considerable performance improvements over SRCNN [52]. FSRCNN [54] achieves high computational efficiency without compromising restoration quality through a redesigned architecture of SRCNN. VDSR [55] addressed the challenge of handling multiscale images within a unified framework by incorporating residual learning, gradient cropping, and an increased number of network layers. EDSR [56] achieves superior performance by streamlining the model architecture, eliminating redundant modules from the conventional ResNet framework. For remote sensing images, LGCNet [57] stands as the pioneering CNN-based model for SR, introducing the concept of local and global contrast features to enhance the preservation of details and clarity in the reconstructed images. Haut et al. [58] coordinate several different improvements in network design to achieve the most advanced performance on

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the RSISR task. A novel single-path feature reuse approach and a second-order learning mechanism are proposed by Dong et al. [59], which aim to effectively utilize both small and large difference features. Although these methods have achieved impressive results, the limited receptive field of CNNs cannot capture the long-range dependencies between pixels, thereby limiting their performance. Zhang et al. [31] propose a hybrid high-order attention network (MHAN) for RSISR, effectively utilizing hierarchical features through high-order attention (HOA) modules of varying orders applied to feature maps from different frequency bands. Inspired by the lattice structure, Wang et al. [32] develop the Fenet network, which incorporates lightweight lattice blocks (LLBSs) as the fundamental module, thus enhancing the model's expressiveness. Maeda [33] proposes an end-to-end SR network with a deep dictionary (SRDD), explicitly learning an HR dictionary while preserving the advantages of deep learning.

B. Transformer-Based Methods

The Transformer network, initially proposed in 2017 for machine translation tasks [60], has gained popularity in computer vision due to its remarkable performance in image processing [35], [61]. Since its inception, numerous visual models based on the Transformer have been proposed [41], [43], [50], [51], [62]. As an example, Chen et al. [41] introduced the image processing transformer (IPT) as a novel pre-trained model for low-level computer vision tasks. To fully leverage the potential of the transformer, a substantial amount of corrupted image pairs is generated using the ImageNet dataset. The IPT model adapts to diverse image processing tasks through multihead and multitail training, along with the incorporation of contrastive learning techniques. Another well-known image restoration method called Uformer [43] utilizes the locally enhanced window transformer block, which reduces computational requirements by utilizing a non-overlapping window-based self-attention mechanism. Moreover, three skip connection schemes are explored to facilitate efficient information transfer from the encoder to the decoder. Furthermore, the restoration transformer (Restormer) network [51] proposes an effective transformer model that captures remote pixel interactions and is suitable for large image, which is achieved through key design choices in the building blocks, including multihead attention and feedforward networking. Efficient SR transformer (ESRT) [62] presents a hybrid model that combines a lightweight CNN backbone (LCB) with a lightweight transformer backbone (LTB) can dynamically adjusts the feature map size, achieving competitive results with low computational cost. SwinIR [42] introduced a robust baseline model for image restoration that utilizes the Swin Transformer architecture. OmniSR [44] proposes the Omni Self-Attention paradigm for simultaneous spatial and channel interactions, effectively capturing potential correlations across all axes. To enhance the reconstruction of input pixels, Chen et al. [45] introduce the novel hybrid attention transformer (HAT), which combines channel attention and self-attention schemes, harnessing their complementary advantages. Additionally, Choi et al. [46] pioneer the use of N-Gram context in low-level vision with Transformers. In the context of RSISR, TransENet [50] proposes a multilevel enhancement architecture based on the Transformer framework, which can be integrated with the conventional SR framework to effectively merge multiscale high- and low-dimensional features.

III. Methodology

In this section, we introduce the proposed SPIFFNet for RSISR. The overall framework of SPIFFNet is presented in Section III-A and the SPIFFNet group is carefully discussed in Section III-B. Furthermore, in Section III-C, we will provide a concise overview of the implementation details.

A. Overview of SPIFFNet

In this section, we introduce the framework of our method, as shown in Fig. 4. Given an input image $I_{\mathrm{LR}} \in \mathbb{R}^{H\times W\times 3}$, where H and W are the image height and width. Then, the input I_{LR} undergoes a transformation into the feature space through a 3×3 convolutional layer

$$F_0 = \mathrm{Conv}(I_{\mathrm{LR}}) \tag{1}$$

where the Conv denotes 3×3 convolution and the $F_0 \in \mathbb{R}^{H\times W\times C}$ represents the shallow features.

Then, several SPIFFNet groups, each of which involves cross-spatial pixel integration attention block (CSPIB), local spatial attention block (LSAB), local 3×3 convolution, and cross-stage feature fusion block (CSFFB), are set up after the convolutional layer for deep feature extraction. We extract deep feature $F_{\mathrm{DF}} \in \mathbb{R}^{H\times W\times C}$ from F_0 as

$$F_{\mathrm{DF}} = H_{\mathrm{DF}}(F_0) \tag{2}$$

where H_{DF} represents the deep feature extraction module which contains K SPIFFNet groups. Specifically, the intermediate features $F_1, F_2, \ldots, F_K$ and the final deep feature F_{DF} are extracted sequentially

$$F_i = H_i(F_{i-1}), \quad i = 1, 2, \ldots, K \tag{3}$$

where H_i denotes the ith SPIFFNet group.

Finally, the deepest features F_{DF} are reconstructed using a 3×3 convolutional layer and pixel-shuffle upsampling operations [63] to generate SR image I_r. In addition, a bilinear interpolation of the LR image I_b is incorporated in the summation process to aid in the recovery process for the SR output I_r

$$I_{SR} = I_r + I_b. \tag{4}$$

We train the proposed model using the L1 loss function. The loss function is obtained by comparing the LR images I_{LR} with their corresponding HR reference images I_{HR}

$$L_{(\theta)} = \frac{1}{N}\sum_{i=1}^{N}\left\| I_{\mathrm{HR}}^{(i)} - I_{SR}^{(i)} \right\|_1 \tag{5}$$

where θ represents the parameters of the SPIFFNet, and N denotes the number of training samples.

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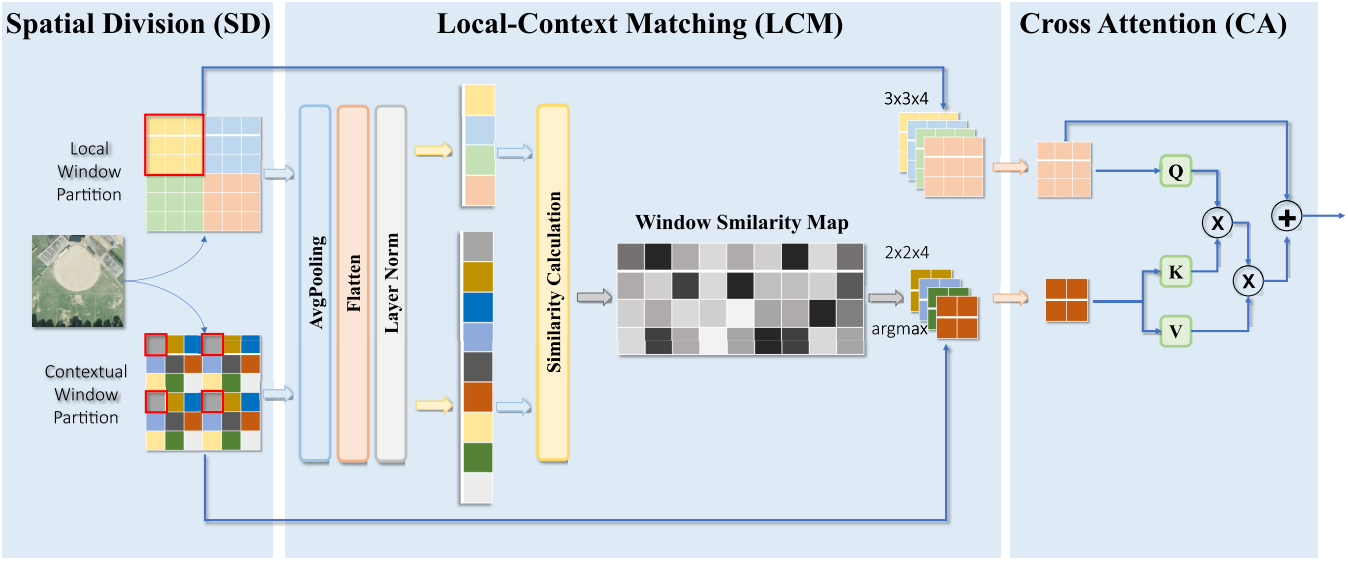


Fig. 5. Illustration of CSPIA. CSPIA consists of three main steps: SD, LCM, and CA. SD is responsible for obtaining local windows and contextual windows. LCM is used to obtain the most matched contextual window for each pair of local window. The most similar contextual window is selected to conduct CA with corresponding local window. ⊗ and ⊕ are matrix multiplication and element-wise addition operations, respectively.

B. SPIFFNet Group

In this section, we introduce the SPIFFNet group, a crucial component of our SPIFFNet model. Each SPIFFNet group consists of four components: CSPIB, LSAB, local 3×3 convolution, and CSFFB. The CSPIB expands the model's receptive field, allowing it to capture long-range dependencies and contextual information from the input feature maps. The LSAB is responsible for capturing the correlations of local spatial information. Local 3×3 convolution deals with local details in a fine-grained manner. CSFFB is designed to adaptive integration of information from the previous stage according to the needs of the characteristics of the current stage rather than treating them equally. This combination of local and contextual information, along with the cross stage adaptive information fusion, enables the model to capture complex spatial dependencies and contextual information effectively.

1) Cross-Spatial Pixel Integration Block: Previous methods often overlooked the interaction between local and contextual features. The CSPIB is designed to expand the local spatial window to capture more context information, as shown in Fig. 4(c). CSPIB contains two sequential modules: the CSPIA is designed to capture contextual information for local windows and the multi-layer perceptron (MLP) module for feature projection.

2) Cross-Spatial Pixel Integration Attention (CSPIA): In this section, our objective is to expand the receptive field of local windows, enabling them to capture context information from the input feature maps. Previous studies have demonstrated that SR networks with a wider effective receptive field achieve superior performance [65]. The challenge lies in enabling the network to model global connectivity while preserving computational efficiency. Due to the fixed partitioning of windows at the layer level, there are no direct connections between windows. One straightforward approach is to exhaustively combine the information from every window pair. However, this approach is unnecessary and inefficient since many windows are irrelevant and uninformative. Additionally, redundant interactions may introduce noise that impairs the model's performance. Based on these observations, we introduce an innovative technique called CSPIA, where each local window adaptively integrates pixels with the most correlated contextual window. Specifically, as shown in Fig. 5, the CSPIA consists of three steps: spatial division (SD), local-context matching (LCM), and cross attention (CA). Through SD, we split the feature map into two parts: local windows and contextual windows. In the case of local windows, adjacent embeddings of size $G \times G$ are grouped together. An example is illustrated in Fig. 2 with $G = 4$. In the case of contextual windows, where the input size is $S \times S$, the feature map is sampled at a fixed interval I. Fig. 2 demonstrates an example with $I = 2$, where embeddings with a red border belong to a window. The height or width of the group for contextual windows is calculated as $G = S/I$. All windows can be processed in parallel, after which the outputs are pasted to their original location in window aggregation module. It is worthwhile to note that such cropping strategy is adaptive to arbitrary input size, which means no padding pixels are needed. Then, local windows and contextual windows are spatially pooled into 1-D tokens. These tokens encode the characteristic of the windows, which are later used for similarity calculation and window matching. This process can be expressed as

$$\bar{X}_i = \arg\max_{X_j} L(X_i)^T L(X_j),\ j \neq i \tag{6}$$

where $\bar{X}_i$ is the best-matching contextual window with current local window X_i, and $L(\cdot)$ is the average pooling function along spatial dimension followed by flatten operation and layer normalization. Since the *argmax* operation is non-differentiable, we replace it with Gumbel-Softmax operation [66] during training so as to make it possible to

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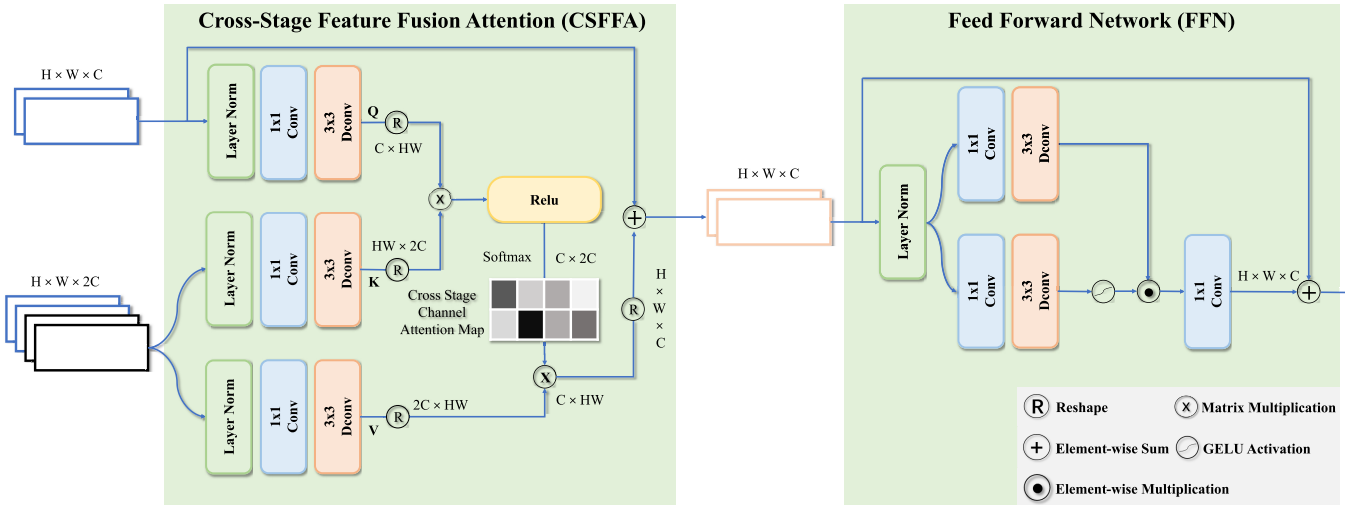


Fig. 6. Illustration of CSFFB. It consists of two parts: CSFFA and FFN. CSFFA facilitates adaptive cross-stage feature fusion by employing cross-stage cross-attention, and FFN is used for feature transformation.

train end-to-end. After that, pixel information of $\bar{X}_i$ is fused into X_i via cross-attention

$$X_i = CA\left(X_i, \bar{X}_i\right). \tag{7}$$

As illustrated in Fig. 5, CA works in a similar way to the standard self-attention [35], but the key and value are calculated using $\bar{X}_i$. As a result, CSPIA can enable contextual pixel integration while introducing little computational overhead.

3) Local Spatial Attention Block: As shown in Fig. 4(d), LSAB adopts the standard multihead self-attention (MSA) paradigm [35], with two modifications. First, LSAB operates at the window level instead of the image level. Second, positional embedding is omitted due to the introduction of the convolutional layer, which implicitly learns positional relationships and enhances the network's efficiency and conciseness. LSAB is designed to model local-range dependencies within a window, facilitating the comprehensive utilization of contextual information.

Specifically, for feature $X \in \mathbb{R}^{P^2 \times C}$, the corresponding *query*, *key*, and *value* matrices $Q \in \mathbb{R}^{P^2 \times d}$, $K \in \mathbb{R}^{P^2 \times d}$, $V \in \mathbb{R}^{P^2 \times C}$ are computed as

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V \tag{8}$$

where the weight matrices W_Q, W_K, and W_V are shared across windows, and P is the window size. By comparing the similarity between Q and K, we obtain an attention map of size $\mathbb{R}^{P^2 \times P^2}$ and multiply it with V. Overall, the calculation of MSA can be expressed as

$$\text{MSA}(X) = \text{softmax}\left(QK^T/\sqrt{d}\right)V. \tag{9}$$

Here, $\sqrt{d}$ is used to control the magnitude of QK^T before applying the softmax function.

Similar to the conventional transformer layer [35], the MLP is employed after MSA module to further transform features. MLP contains two fully-connected layers, and one Gaussian error linear unit (GELU) nonlinearity is applied after the first linear layer.

4) Local 3 × 3 Convolution: By adding a local 3 × 3 convolutional layer after feature extraction, the Transformer-based network is infused with the inherent inductive bias of convolution operations. This enhances the foundation for aggregating shallow and deep features in subsequent stages.

5) Cross-Stage Feature Fusion Block: Skip connections are commonly used to propagate shallow features to deeper layers. However, the long-term information from the shallow stages tends to be attenuated. Although the shallow features can be reused through skip connections, they are treated indiscriminately with the deep features across different stages, thereby impeding the representational capacity of CNNs. To address this concern, we introduce CSFFB, as illustrated in Fig. 4(e). CSFFB comprises two consecutive components: the CSFFA for adaptive feature fusion across stages and the FFN for feature transformation.

6) Cross-Stage Feature Fusion Attention: Fig. 6 demonstrates the operation of the CSFFA, which calculates attention scores at the channel level using feature maps cross-stages. These scores are then applied to the feature maps, enabling the weighting of each channel's contribution and the integration of features from previous stages with the current input. This process promotes the fusion of channel information across stages, facilitating the model to capture both low-level details and high-level contextual information. As a result, the approach enables more precise and effective image SR.

Specifically, the features of the current stage, denoted as $X_{\text{cur}} \in \mathbb{R}^{H \times W \times C}$, and the previous stage, denoted as $X_{\text{pre}} \in \mathbb{R}^{H \times W \times C}$, are concatenated along the feature dimension to obtain $Y \in \mathbb{R}^{H \times W \times 2C}$. Our CSFFA then gets *query*, *key*, and *value* projections: $Q = W_d^Q W_p^Q X_{\text{cur}}$, $K = W_d^K W_p^K Y$, and $V = W_d^V W_p^V Y$. Where $W_p^{(\cdot)}$ denotes the 1 × 1 point-wise convolution, and $W_d^{(\cdot)}$ represents the 3 × 3 depth-wise convolution. Subsequently, we reshape the Q and K to facilitate their dot–product interaction, resulting in a cross-stage channel attention map A with dimensions $\mathbb{R}^{C \times 2C}$. The CSFFA process

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TABLE I

MEAN PSNR (DB) AND SSIM OVER THE UCMERCED TEST DATASET AND THE RSSCN7 TEST DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Dataset	Scale	Metric	SRCNN [52]	VDSR [55]	DCM [58]	MHAN [31]	HSENet [64]	TransENet [50]	SRDD [33]	FENet [32]	OmniSR [44]	SPIFFNet (Ours)
UCMerced	×2	PSNR	32.94	33.87	33.65	33.92	34.22	34.05	34.12	33.95	34.16	34.67
		SSIM	0.9170	0.9274	0.9274	0.9283	0.9327	0.9294	0.9303	0.9284	0.9303	0.9351
		SCC	0.5770	0.6196	0.6291	0.6242	0.6341	0.6275	0.6331	0.6243	0.6326	0.6519
		SAM	0.0580	0.0519	0.0507	0.0518	0.0500	0.0511	0.0507	0.0518	0.0506	0.0476
	×3	PSNR	28.97	29.75	29.86	29.94	30.04	29.90	29.92	29.80	29.99	30.43
		SSIM	0.8132	0.8346	0.8393	0.8391	0.8433	0.8397	0.8411	0.8379	0.8403	0.8516
		SCC	0.3404	0.3941	0.4025	0.4034	0.4131	0.3988	0.4084	0.3941	0.4073	0.4339
		SAM	0.0904	0.0829	0.0820	0.0816	0.0806	0.0816	0.0815	0.0826	0.0810	0.0771
	×4	PSNR	26.92	27.54	27.60	27.63	27.75	27.78	27.67	27.59	27.80	28.09
		SSIM	0.7286	0.7522	0.7556	0.7581	0.7611	0.7635	0.7609	0.7538	0.7637	0.7733
		SCC	0.2156	0.2589	0.2610	0.2649	0.2692	0.2701	0.2718	0.2568	0.2779	0.2969
		SAM	0.1128	0.1055	0.1051	0.1043	0.1034	0.1029	0.1047	0.1053	0.1027	0.0993
RSSCN7	×2	PSNR	29.83	30.04	30.03	30.06	30.15	30.08	30.05	30.05	30.11	30.26
		SSIM	0.7954	0.8027	0.8024	0.8036	0.8070	0.8040	0.8051	0.8033	0.8052	0.8095
		SCC	0.2816	0.2967	0.2979	0.2996	0.3056	0.2984	0.2993	0.2991	0.3023	0.3155
		SAM	0.1042	0.1018	0.1019	0.1016	0.1006	0.1013	0.1014	0.1016	0.1010	0.0996
	×3	PSNR	27.77	27.94	27.96	28.00	28.02	28.02	27.96	27.97	28.04	28.15
		SSIM	0.6936	0.7010	0.7027	0.7045	0.7067	0.7054	0.7052	0.7031	0.7061	0.7105
		SCC	0.1368	0.1495	0.1524	0.1551	0.1572	0.1532	0.1552	0.1524	0.1584	0.1672
		SAM	0.1326	0.1303	0.1301	0.1296	0.1292	0.1292	0.1300	0.1300	0.1290	0.1278
	×4	PSNR	26.64	26.75	26.79	26.79	26.82	26.81	26.74	26.80	26.85	26.93
		SSIM	0.6278	0.6336	0.6363	0.6360	0.6378	0.6373	0.6364	0.6367	0.6388	0.6425
		SCC	0.0754	0.0825	0.0867	0.0850	0.0867	0.0845	0.0842	0.0871	0.0898	0.0935
		SAM	0.1513	0.1495	0.1490	0.1491	0.1485	0.1485	0.1495	0.1487	0.1480	0.1471

can be defined as

$$X_{\text{out}} = W_p \text{Attn}(Q, K, V) + X_{\text{cur}}$$
$$\text{Attn}(Q, K, V) = V \cdot \text{Softmax}(\text{Relu}(Q \cdot K/\alpha)) \quad (10)$$

where X_{cur} and X_{out} are the input and output feature maps, respectively. $Q \in \mathbb{R}^{C\times HW}$, $K \in \mathbb{R}^{HW\times 2C}$, and $V \in \mathbb{R}^{2C\times HW}$ matrices are obtained by the $X_{\text{cur}} \in \mathbb{R}^{H\times W\times C}$ and $Y \in \mathbb{R}^{H\times W\times 2C}$, respectively. To enhance feature control and promote the development of sophisticated image attributes, we introduce a rectified linear unit (ReLU) nonlinearity function before the softmax normalization. The ReLU nonlinearity function applies sparse constraints to the cross-stage attention map promotes the model's focus on the most informative regions and mitigates the influence of noisy or irrelevant features.

To transform features, we use two parallel 1×1 convolutions and 3×3 convolutions to the feature map. Subsequently, a SimpleGate activation function [67] multiplies one of the branches to regulate the flow of complementary features and facilitate feature transformation. To be specific, when provided with an input tensor $X \in \mathbb{R}^{H\times W\times C}$, the FFN can be expressed as

$$\hat{X} = W_p^{0\text{Gating}}(X) + X$$
$$\text{Gating}(X) = \phi(W_d^1 W_p^1(LN(X))) \odot W_d^2 W_p^2(LN(X)) \quad (11)$$

where the symbol $\odot$ denotes element-wise multiplication, ϕ refers to the GELU nonlinearity function, and LN denotes layer normalization [68].

C. Implementation Details

This article focuses on RSISR at three magnification factors: ×2, ×3, and ×4. During the training, we randomly sample LR remote sensing images and their corresponding HR reference windows as 48×48 windows. To augment the training samples, we apply random rotations (90°, 180°, and 270°) and horizontal flipping. The proposed SPIFFNet consists of ten blocks, each of which is designed with a local window size and a contextual window size of 16 and a feature dimension of 64. Additionally, the SPIFFNet employs four attention heads. Further details and experimental analyses are presented in Section IV.

We employ the Adam optimizer [69] for model optimization, setting $\beta_1 = 0.9$, $\beta_2 = 0.99$, and $\varepsilon = 10^{-8}$. We set the initial learning rate to 4×10^{-4} and the mini-batch size to 8. We train the model for a total of 2000 epochs, gradually

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TABLE II

MEAN PSNR (DB) AND SSIM OVER THE AID TEST DATASET AND WHU-RS19 DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Dataset	Scale	Metric	SRCNN [52]	VDSR [55]	DCM [58]	MHAN [31]	HSENet [64]	TransENet [50]	SRDD [33]	FENet [32]	OmniSR [44]	SPIFFNet (Ours)
AID	×2	PSNR	34.65	35.11	35.35	35.56	35.50	35.40	35.33	35.33	35.50	35.66
		SSIM	0.9290	0.9340	0.9366	0.9390	0.9383	0.9372	0.9367	0.9364	0.9383	0.9397
		SCC	0.6044	0.6181	0.6407	0.6641	0.6626	0.6538	0.6373	0.6390	0.6523	0.6679
		SAM	0.0576	0.0544	0.0531	0.0520	0.0524	0.0530	0.0531	0.0533	0.0522	0.0515
	×3	PSNR	30.55	31.17	31.31	31.55	31.49	31.50	31.38	31.33	31.53	31.70
		SSIM	0.8372	0.8511	0.8561	0.8603	0.8588	0.8588	0.8564	0.8550	0.8596	0.8631
		SCC	0.3538	0.3800	0.3946	0.4098	0.4053	0.4067	0.3984	0.3955	0.4081	0.4174
		SAM	0.0891	0.0836	0.0822	0.0801	0.0806	0.0806	0.0817	0.0823	0.0803	0.0789
	×4	PSNR	28.48	28.99	29.20	29.39	29.32	29.44	29.21	29.16	29.19	29.54
		SSIM	0.7565	0.7753	0.7826	0.7892	0.7867	0.7912	0.7835	0.7812	0.7829	0.7938
		SCC	0.2114	0.2427	0.2679	0.2825	0.2765	0.2884	0.2695	0.2651	0.2636	0.2962
		SAM	0.1119	0.1055	0.1032	0.1008	0.1017	0.1002	0.1030	0.1037	0.1033	0.0995
WHU-RS19	×2	PSNR	30.19	30.22	30.18	30.19	30.17	30.22	30.17	30.18	30.18	30.24
		SSIM	0.8262	0.8273	0.8258	0.8258	0.8255	0.8267	0.8256	0.8257	0.8257	0.8267
		SCC	0.3035	0.3044	0.3068	0.3082	0.3057	0.3102	0.3051	0.3056	0.3071	0.3086
		SAM	0.0858	0.0856	0.0861	0.0859	0.0861	0.0856	0.0860	0.0860	0.0860	0.0858
	×3	PSNR	27.79	27.80	27.78	27.81	27.79	27.86	27.78	27.79	27.80	27.88
		SSIM	0.7307	0.7311	0.7299	0.7309	0.7305	0.7325	0.7297	0.7302	0.7305	0.7329
		SCC	0.1779	0.1814	0.1828	0.1851	0.1840	0.1880	0.1824	0.1829	0.1845	0.1894
		SAM	0.1142	0.1142	0.1143	0.1139	0.1141	0.1132	0.1143	0.1141	0.1140	0.1135
	×4	PSNR	26.35	26.36	26.34	26.37	26.37	26.42	26.36	26.36	26.34	26.44
		SSIM	0.6601	0.6609	0.6606	0.6618	0.6615	0.6639	0.6607	0.6606	0.6605	0.6643
		SCC	0.1080	0.1120	0.1163	0.1190	0.1183	0.1227	0.1166	0.1162	0.1153	0.1232
		SAM	0.1354	0.1352	0.1353	0.1349	0.1349	0.1341	0.1351	0.1351	0.1351	0.1343

TABLE III

QUANTITATIVE COMPARISON OF PARAMETERS AND FLOPS ON THE UCMERCED DATASET FOR ×4 SR

Method	Params	FLOPS	PSNR
SRCNN [52]	69K	4.53G	26.92dB
VDSR [55]	671K	44.03G	27.54dB
LGCNet [57]	193K	0.44G	27.02dB
DCM [58]	2170K	13.00G	27.60dB
MHAN [31]	11500K	67.88G	27.63dB
HSENet [64]	5430K	19.20G	27.75dB
TransENet [50]	37460K	21.44G	27.78dB
SRDD [33]	9340K	6.46G	27.67dB
FENet [32]	366K	1.44G	27.59dB
OmniSR [44]	793K	3.15G	27.80dB
SPIFFNet (Ours)	1750K	7.32G	28.09dB

TABLE IV

ABLATION EXPERIMENTS OF COMPONENT ANALYSIS BY SPIFFNET ON THE UCMERCED-×4

CSPIA	CSFFA	Param	FLOPS	PSNR	SSIM
✗	✗	1.67M	5.15G	28.009	0.7711
✗	✓	1.75M	7.00G	28.039	0.7718
✓	✗	1.67M	5.47G	28.077	0.7723
✓	✓	1.75M	7.32G	28.086	0.7732

reducing the learning rate to 5×10^{-7} at epoch 2000 using the cosine annealing schedule [70].

IV. EXPERIMENTAL RESULTS AND ANALYSES

A. Experimental Datasets and Metrics

1) *Datasets:* In this study, we utilize four publicly available remote sensing datasets: UCMecred [71], RSSCN7 [72], Aerial Image Dataset (AID) [73], and WHU-RS19 [74]. The UCMerced dataset consists of 21 classes, each containing 100 remote sensing scene images with dimensions of 256 × 256 pixels. We divide this dataset into two subsets for training and testing purposes, each comprising 1050 images. The RSSCN7 dataset comprises 2800 images distributed among seven different classes, each with a resolution of 400 × 400 pixels. Following the partitioning approach used for the UCMerced dataset, we split the RSSCN7 dataset into balanced halves for training and testing, with 1400 samples in each subset. Additionally, 20% of the training set is reserved for validation. The AID dataset consists of 10 000 images representing 30 classes of remote sensing scenes, each having dimensions of 600 × 600 pixels. For the AID dataset, we randomly assign 80% of the total dataset as the training

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TABLE V
ABLATION EXPERIMENTS OF HYPERPARAMETER ANALYSIS BY SPIFFNET ON THE UCMERCED-×4

Blocks Numebr	Heads Number	Dim Number	Params	FLOPS	PSNR
5	4	64	1.07M	4.54G	29.27dB
6			1.20M	5.10G	28.00dB
7			1.34M	5.66G	28.03dB
8			1.48M	6.21G	28.07dB
9			1.62M	6.77G	28.08dB
10			1.75M	7.32G	28.09dB
10	1	64	1.75M	7.10G	28.03dB
	2		1.75M	7.17G	28.06dB
	4		1.75M	7.32G	28.09dB
	8		1.75M	7.62G	28.09dB
	16		1.75M	8.22G	28.03dB
10	4	32	0.45M	2.13G	27.85dB
		48	0.99M	4.30G	28.05dB
		64	1.75M	7.32G	28.09dB
		80	2.73M	11.18G	28.10dB
		96	3.92M	15.89G	28.10dB

TABLE VI
MEAN PSNR(DB) OF EACH CLASS FOR UPSCALING FACTOR FOUR ON THE UCMERCED TEST DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Class Name	MHAN [31]	HSENet [64]	TransENet [50]	OmniSR [44]	SPIFFNet (Ours)
agricultural	26.02	26.16	26.04	26.00	26.28
airplane	27.56	27.65	27.75	27.76	28.08
baseball diamond	33.10	33.14	33.09	33.17	33.27
beach	35.48	35.51	35.50	35.52	35.65
buildings	26.23	26.37	26.37	26.50	26.89
chaparral	24.85	24.89	24.94	24.95	25.06
dense residential	26.48	26.58	26.62	26.72	26.99
forest	26.63	26.66	26.64	26.67	26.74
freeway	28.15	28.38	28.27	28.25	28.66
golf course	34.97	35.03	35.05	35.09	35.15
harbor	21.60	21.85	22.01	21.96	22.48
intersection	26.62	26.78	26.79	26.85	27.19
medium residential	25.99	26.11	26.19	26.23	26.52
mobilehomepark	22.98	23.14	23.28	23.39	23.76
overpass	25.80	26.03	25.87	25.86	26.35
parking lot	21.56	21.71	21.94	21.92	22.61
river	27.53	27.60	27.55	27.60	27.65
runway	29.39	29.48	29.92	29.66	30.00
sparse residential	29.12	29.15	29.17	29.23	29.42
storage tanks	30.43	30.54	30.44	30.52	30.76
tennis court	29.81	29.95	29.87	30.02	30.30
avg	27.63	27.75	27.78	27.80	28.09

set, while the remaining images are allocated for testing. The WHU-RS19 dataset contains 1005 images distributed among 19 different classes, with each image having a size of 600 × 600 pixels in RGB color space. All images in the WHU-RS19 dataset serve as a supplement to the AID test set, allowing for a better evaluation of the SR performance and generalization ability of different models.

2) *Metrics:* We utilize peak signal-to-noise ratio (PSNR), structural similarity index measure (SSIM) [75], spatial correlation coefficient (SCC) [76], and spectral angle mapper (SAM) [77] as the evaluation metrics for RSISR. The assessment of all SR results is conducted on the RGB channels. Higher PSNR, SSIM, SCC, and lower SAM values indicate better-reconstructed image quality.

B. Ablation Studies

Before discussing our experimental results, as shown in Tables I–III, we first verify the validity of the proposed SPIFFNet. We conducted experiments in this section to validate the components of our method. All experiments were performed using the same experimental setup, with the UCMereced dataset, and a uniform magnification factor of 4 was applied. We start with a naive baseline by removing CSFFA and replacing CSPIA with ordinary multihead attention. Then, we add CSPIA and CSFFA to the baseline, respectively. At last, both components are employed to compose our final version of the method. Table IV displays the quantitative performance results for ×4 SR on the UCMerced dataset. In comparison to the baseline results, both CSPIA and CSFFA contribute to performance improvements. As a result of incorporating these two modules, the model achieves an additional performance enhancement of 0.077 dB. Subsequently, we will delve into a more detailed explanation of the effects of CSPIA and CSFFA.

1) Effects of CSPIA: To verify that CSPIA can improve the performance of RSISR by integrating context information into local windows, we replace CSPIA with vanilla MSA that integrates pixels only within a local window for comparison. As shown in Table IV, the inclusion of CSPIA results in consistent improvements in PSNR and SSIM, signifying that the introduction of context information into local windows leads to enhanced reconstruction quality.

To better understand the performance improvement by CSPIA, we utilize LAM [65] to visualize the effective receptive field of an input window. As depicted in Fig. 7, the window derives benefits from a broader range of relevant global pixels through the utilization of CSPIA. Consequently, the model that integrates CSPIA extends its coverage over the pixels in use, leading to superior reconstruction results, and demonstrating the efficacy of the proposed CSPIA in enhancing RSISR performance.

2) Effects of CSFFA: To validate the efficacy of the CSFFA, we constructed a model variant where we excluded CSFFA from all the blocks. The results are reported in Table IV. The model that integrates CSFFA exhibits a marked performance improvement compared to the baseline model. In contrast to conventional transformer-based methods that treat shallow and deep features equally through skip connections, the CSFFA leverages an adaptive feature fusion mechanism from the preceding stage to enhance feature representations, resulting in significantly higher PSNR and SSIM values. This provides strong evidence of the effectiveness of CSFFA.

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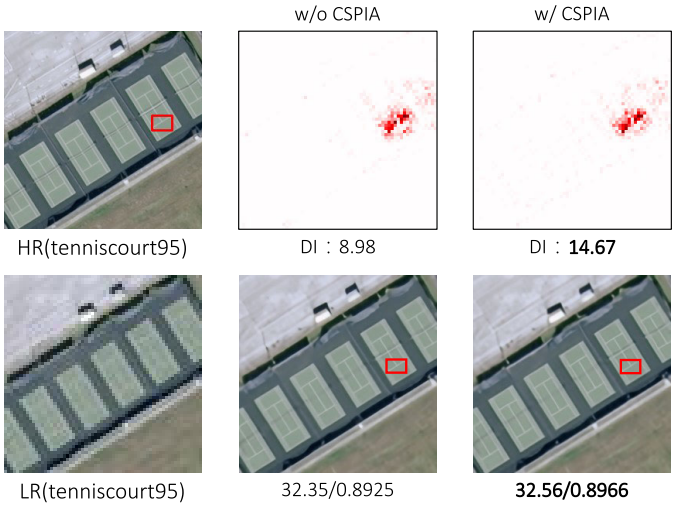


Fig. 7. LAM [65] comparison between the full SPIFFNet (w/CSPIA) and the variant without CSPIA (w/o CSPIA) for ×4 SR. The first column displays the low remote sensing resolution image alongside its corresponding high remote sensing resolution image. The second and third columns present SPIFFNet and its variants, respectively, along with their corresponding effective receptive fields. The quantitative diffusion index (DI) is utilized to reflect the range of involved pixels, with a higher DI value indicating a wider attention range.

TABLE VII

MEAN PSNR(DB) OF EACH CLASS FOR UPSCALING FACTOR 4 ON THE RSSCN7 TEST DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Class Name	MHAN [31]	HSENet [64]	TransENet [50]	OmniSR [44]	SPIFFNet (Ours)
grass	32.28	32.28	32.26	32.31	32.45
field	31.48	31.50	31.48	31.54	31.68
industry	22.99	23.06	23.04	23.08	23.15
river lake	28.78	28.81	28.79	28.83	28.99
forest	26.61	26.63	26.64	26.66	26.67
resident	21.90	21.94	21.97	21.98	22.00
parking	23.47	23.53	23.49	23.53	23.56
avg	26.79	26.82	26.81	26.85	26.93

3) Hyperparameter Analysis: The hyperparameter analysis is presented in Table V, encompassing the number of attention heads, blocks, and channel dimensions for SPIFFNet. Based on the results, we determine the optimal trade-off between performance and model latency. Specifically, we set the number of attention heads to 4, the number of blocks to 10, and the channel dimension to 64.

C. Comparisons With Other Methods

This section presents a comparative analysis of the proposed method with several deep learning-based SR methods, namely SRCNN [52], VDSR [55], LGCNet [57], DCM [58], MHAN [31], HSENet [64], TransENet [50], SRDD [33], FENet [32], and OmniSR [44].

1) Parameters and FLOPs: Table III provides a comparison of model size and computational speed among these methods. While LGCNet [57], DCM [58], SRDD [33], FENet [32], and OmniSR [44] perform well in FLOPs, they manifest

TABLE VIII

MEAN PSNR(DB) OF EACH CLASS FOR UPSCALING FACTOR 4 ON THE AID TEST DATASET. THE BEST AND SECOND BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Class Name	MHAN [31]	HSENet [64]	TransENet [50]	OmniSR [44]	SPIFFNet (Ours)
airport	29.20	29.12	29.26	28.98	29.34
bare land	36.36	36.34	36.38	36.23	36.48
baseball field	31.57	31.49	31.63	31.33	31.48
beach	32.62	32.60	32.66	32.48	32.78
bridge	31.65	31.55	31.70	31.40	31.87
center	28.03	27.91	28.09	27.73	28.21
church	24.50	24.43	24.53	24.27	24.63
commercial	27.96	27.90	28.00	27.80	28.07
dense residential	25.08	25.02	25.17	24.88	25.20
desert	39.50	39.47	39.55	39.31	39.58
farmland	34.65	34.59	34.67	34.45	34.79
forest	28.52	28.54	28.59	28.49	28.62
industrial	27.18	27.09	27.24	26.94	27.32
meadow	33.00	32.97	33.00	32.93	33.14
medium residential	28.48	28.41	28.50	28.26	28.57
mountain	29.24	29.22	29.30	29.21	29.33
park	27.97	27.93	28.04	27.84	28.16
parking	26.35	26.16	26.49	25.70	26.56
playground	30.34	30.19	30.38	29.97	30.64
pond	30.53	30.48	30.58	30.38	30.86
port	26.90	26.80	26.95	26.62	26.99
railway station	28.59	28.52	28.64	28.42	28.74
resort	28.07	28.00	28.13	27.89	28.13
river	31.00	30.97	31.04	30.93	31.11
school	27.18	27.10	27.25	26.98	27.35
sparse residential	26.62	26.60	26.63	26.52	26.68
square	29.38	29.30	29.46	29.14	29.59
stadium	27.40	27.28	27.48	27.13	27.54
storage tanks	26.20	26.12	26.22	26.00	26.26
viaduct	28.19	28.09	28.24	27.95	28.36
avg	29.39	29.32	29.44	29.19	29.54

clear deficiencies in terms of reconstruction accuracy. Additionally, SPIFFNet attained the highest PSNR value when compared to larger models, including DCM [58], MHAN [31], HSENet [64], TransENet [50], and SRDD [33]. This illustrates that SPIFFNet strikes a balance between effectiveness and efficiency by introducing context information and adaptively adjusting cross-stage features.

2) Quantitative Results: The quantitative analyses of the various methods, evaluated using different datasets and zooming factors, are presented in Tables I and II. To ensure consistency, we retrained the comparison methods using open-source code under the same testing conditions, as some results were reported in multiple published articles [58], [78], [79]. In Table I, when considering training on both the UCMerced and RSSCN7 datasets, our proposed SPIFFNet

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Fig. 8. Result comparisons on the UCMerced dataset and RSSCN7 dataset with different methods.

outperforms all other methods. Specifically, SPIFFNet attains the highest PSNR, SSIM, SCC, and SAM values on both datasets, with the PSNR/SSIM/SCC/SAM advantage ranging from +0.43dB/+0.0096/+0.0208/-0.0035 to +0.29dB/+0.0024/+0.0178/−0.0024 over the second-ranked HSENet [64]. Compared with HSENet's emphasis on

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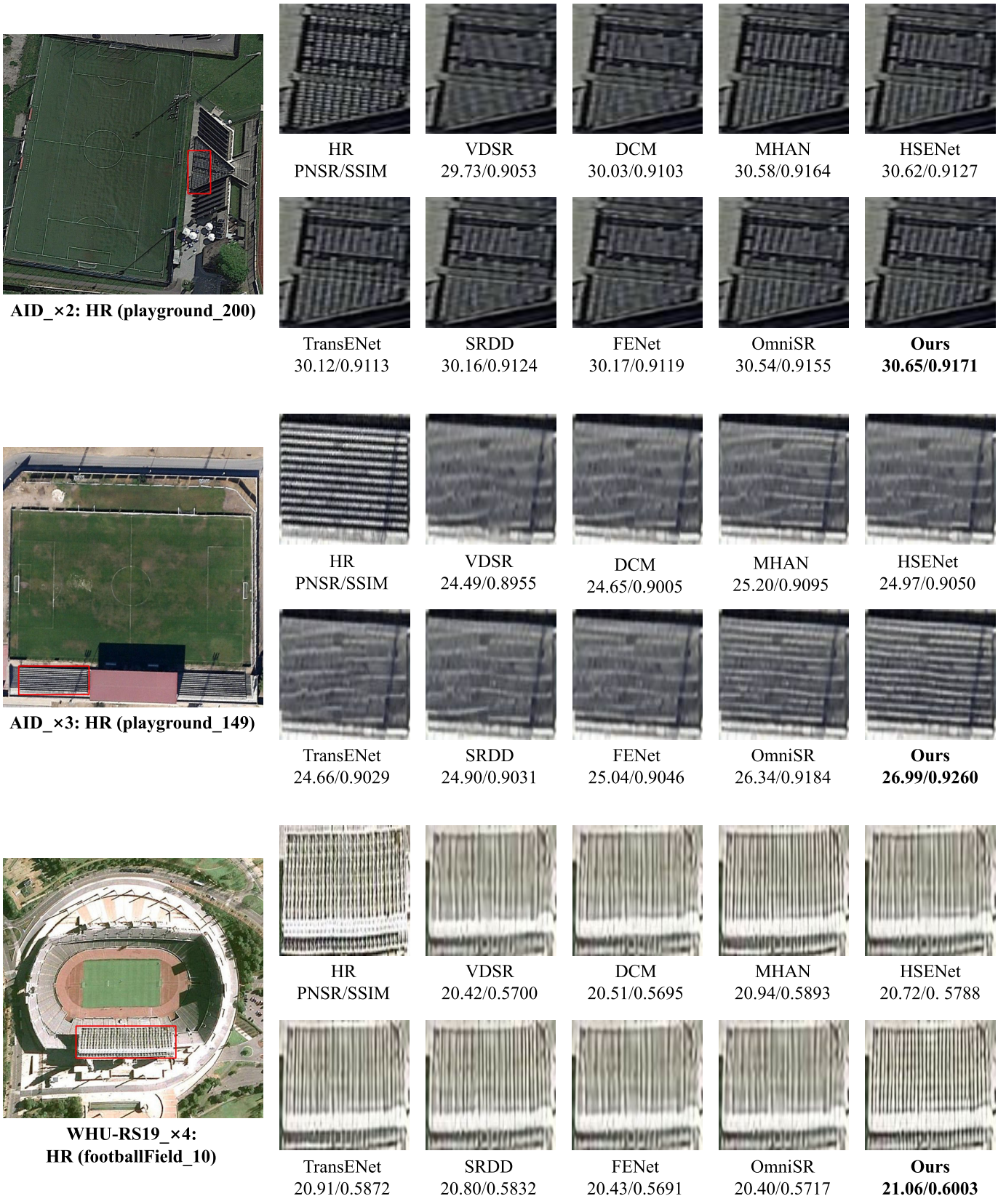


Fig. 9. Result comparisons on the AID dataset and WHU-RS19 dataset with different methods.

investigating mixed-scale feature correlations, SPIFFNet introduces global features by calculating the cross-attention between the local window and its most relevant context window, which can better enhance the model's overall perception of the image. Notably, as demonstrated in Table II, even though SR performance declines on certain

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indices compared to TransENet [50] when training on the larger AID dataset and testing on the WHU-RS19 dataset, SPIFFNet still manages to deliver nearly the best performance across all indices in the AID and WHU-RS19 test datasets. Nevertheless, it is important to note that TransENet [50] has approximately 20 times more parameters and three times more FLOPs compared to SPIFFNet. Compared with TransENet's redundant global attention, SPIFFNet's cross-spatial pixel integration attention and CSFFA achieve a better balance between model performance and size. Overall, our proposed approach consistently maintains competitive performance.

To delve deeper into the reasons behind the observed phenomenon, we conduct a comprehensive discussion on the quantitative performance of different methods across various classes within the same dataset. Since the categories of the AID dataset cover all categories in the WHU-RS19 dataset, the results of the WHU-RS19 are not shown. For intuitive analysis of experimental results, we select the four best-performing methods for comparison with ours. Tables VI–VIII present a summary of the PSNR for each class in the remaining three datasets at an upscale factor of 4. Notably, SPIFFNet excels in scenarios where consistency and relevance play crucial roles, such as "parking lot" (+0.67dB in UCMerced), "river lake" (+0.67dB in RSSCN7), and "pond" (+0.28dB in UCMerced), which demand effective utilization of context information. The consistent superior performance of SPIFFNet across various scenarios further substantiates the effectiveness of our approach.

3) Qualitative Results: To gain deeper insights into the reasons behind the observed phenomenon, we present qualitative visualization results to highlight the unique characteristics of each method. In Fig. 8, various super-resolved examples from both the UCMerced dataset and RSSCN7 dataset are presented. These examples encompass scenes such as "field," "river lake," and "overpass." Similarly, Fig. 9 exhibits examples from the AID dataset and WHU-RS19 dataset, including scenes like "playground" and "football field." When compared to other methods, SPIFFNet generates super-resolved images that closely mirror the ground-truth HR images, displaying well-defined edges and contours. For instance, in the examples from the RSSCN7 and UCMerced datasets shown in Fig. 8, SPIFFNet restores a greater amount of high-frequency detail compared to OmniSR [44] and TransENet [50], which have the second-highest PSNR and SSIM values. This demonstrates that in comparison to OmniSR [44], which models pixel interactions in both spatial and channel dimensions, and TransENet [50], which models global pixel interactions, SPIFFNet is capable of capturing a greater amount of information concerning global local detail recovery by incorporating context information into local windows while simultaneously adapting cross-stage features along the channel dimensions. Likewise, in the examples from the AID and WHU-RS19 datasets depicted in Fig. 9, the reconstruction results from SPIFFNet exhibit the highest perceived quality relative to other methods. To demonstrate the practical reconstructive capabilities of our method, in line with [64] and [80], we selected real-world remote sensing images from GaoFen-1, GaoFen-2, and Google Earth, as depicted in Fig. 10. We use the pre-trained SPIFFNet model on the AID dataset to recover high-frequency details of the real images. Despite differences between these images and the training dataset, SPIFFNet excels in capturing additional details, particularly rich and distinct edge information related to typical remote sensing objects. This observation further demonstrates the effectiveness of our approach.

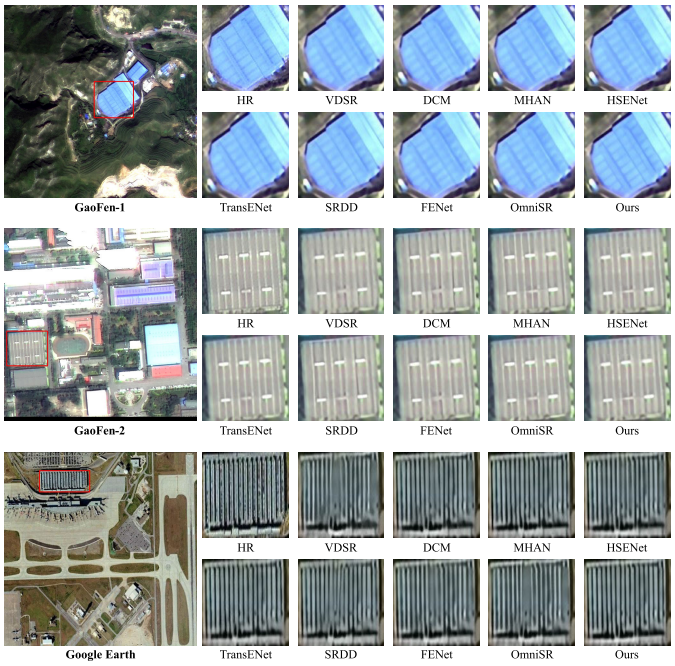


Fig. 10. Visual results on real-world images from GaoFen-1, GaoFen-2, and Google Earth.

V. Conclusion

This article introduces a novel transformer-based method called cross-SPIFFNet for RSISR. The aim of SPIFFNet is to enhance the contextual perception ability of local windows by introducing context information and to improve the representation ability of features by integrating cross-stage features. SPIFFNet consists of two key components: CSPIA, which introduces context information into the reconstruction of local windows to enhance contextual awareness, and CSFFA, which enables adaptive aggregation of features across different stages of the network, resulting in more effective information fusion and superior SR performance. We conducted extensive experiments on benchmark datasets to validate the effectiveness of SPIFFNet. Our proposed SPIFFNet provides a new paradigm for local and contextual information interaction in a single window, which can be applied to other computer vision tasks. In future work, we plan to explore the use of SPIFFNet in other remote sensing applications, such as object detection, semantic segmentation, and image fusion.

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Yuting Lu is currently pursuing the Ph.D. degree in control science and engineering from the Northwestern Polytechnic University, Xi'an, China.

His research interests include image super-resolution and computer vision.



Lingtong Min (Member, IEEE) received the B.S. degree from Northeastern University, Shenyang, China, in 2012, and the Ph.D. degree from Zhejiang University, Hangzhou, China, in 2019.

He is an Associate Professor with Northwestern Polytechnical University, Xi'an, China. His main research interests are computer vision, pattern recognition, and remote sensing image understanding.



Binglu Wang (Member, IEEE) received the Ph.D. degree in control science and engineering from the School of Automation, Northwestern Polytechnic University, Xi'an, China, in 2021.

He is currently a Post-Doctoral Researcher with the Department of Electrical Engineering, Beijing Institute of Technology, Beijing, China. His research interests include computer vision, digital signal processing, and deep learning.



Le Zheng (Senior Member, IEEE) received the B.Eng. degree from Northwestern Polytechnical University (NWPU), Xi'an, China, in 2009, and the Ph.D. degree from the Beijing Institute of Technology (BIT), Beijing, China, in 2015.

He has previously held an academic positions with the Electrical Engineering Department, Columbia University, New York, NY, USA, first as a Visiting Researcher from 2013 to 2014 and then as a Post-Doctoral Research Fellow from 2015 to 2017. From 2018 to 2022, he worked with Aptiv (formerly Delphi), Los Angeles, CA, USA, as a Principal Radar Systems Engineer, leading projects on the next-generation automotive radar products. Since July 2022, he has been a Full Professor with the School of Information and Electronics, BIT. His research interests lie in the general areas of radar, statistical signal processing, wireless communication, and high-performance hardware, and in particular in the area of automotive radar and integrated sensing and communications (ISAC).

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Xiaoxu Wang (Member, IEEE) received the M.S. and Ph.D. degrees from the School of Automation, Harbin Engineering University, Harbin, China, in 2008 and 2010, respectively.

He was as a Post-Doctoral Researcher from 2010 to 2012, and as an Associate Professor from 2013 to October 2018 with the School of Automation, Northwestern Polytechnical University, Xi'an, China, where he is currently a Professor. His main research interests include deep learning, inertial navigation, and nonlinear estimation.



Yongqiang Zhao (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in control science and engineering from the Northwestern Polytechnic University, Xi'an, China, in 1998, 2001, and 2004, respectively.

From 2007 to 2009, he was as a Post-Doctoral Researcher with McMaster University, Hamilton, ON, Canada, and Temple University, Philadelphia, PA, USA. He is currently a Professor with Northwestern Polytechnical University, Xi'an. His research interests include polarization vision, hyperspectral imaging, and pattern recognition.

Le Yang received the B.E. and Ph.D. degrees from Northwestern Polytechnical University, Xi'an, China, in 2016 and 2022, respectively.

He is currently a Post-Doctoral Researcher with the Institute of Artificial Intelligence, Hefei Comprehensive National Science Center, Hefei, China. His research interests include video analysis and weakly supervised learning.



Teng Long (Fellow, IEEE) was born in Fujian, China, in 1968. He received the M.S. and Ph.D. degrees in electrical engineering from the Beijing Institute of Technology, Beijing, China, in 1991 and 1995, respectively.

He was a Visiting Scholar with Stanford University, Stanford, CA, USA, in 1999, and University College London, London, U.K., in 2002. He has been a Full Professor with the Department of Electrical Engineering, Beijing Institute of Technology, since 2000. He has authored or coauthored more than 300 articles. His research interests include synthetic aperture radar systems and real-time digital signal processing, with applications to radar and communication systems.

Dr. Long is a fellow of the Institute of Electronic and Technology and the Chinese Institute of Electronics. He has been a member of the Chinese Engineering Academy since 2021. He was a recipient of many awards for his contributions to research and invention in China.

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